

The Portfolio Scenario:
National Healthcare Resource Allocation – Phase 1
(Data Analysis Project)

Data Analyst: Symone Thompson
Client/Sponsor: National Maternal Health Organization
Target Asset: O1_baseline_structural_audit.csv

Executive Summary & Objective:
The primary goal for this inaugural phase involves the ingestion of raw CDC birth logs into the Google Cloud Platform workspace console to define a technical structural baseline. Through an exhaustive audit of table definitions and field parameters, this initial stage provides healthcare leadership with a high-integrity data ecosystem, ensuring that all future resource allocation findings are derived from a validated and hygienic structural framework.

Purpose:
This foundational stage involves connecting to the public health data ecosystem to investigate the raw record distribution absent any active sorting constraints or specific filters. By establishing this structural mapping, the project can validate data type precision, identify inherent null definitions, and audit primary metrics, ensuring a hygienic framework before proceeding with sophisticated regional logistics allocations.

Scope / Major Project Activities:
The major parts of this project. High-level steps, activities, or stages of the project, and a brief description for each.

Activity	Description
Environment Setup & Data Ingestion:	Access the Google Cloud Platform sandbox, initialize a BigQuery workspace console, and connect to the public <code>sdoh_cdc_wonder_natality</code> data ecosystem. Ingestion of historical CDC Wonder Natality county logs via the Google Cloud Platform sandbox environment.

Schema Assessment & Data Hygiene	Audit the technical structural layout of the <code>county_natality</code> table definitions, verifying fields like <code>Year</code> , <code>County_of_Residence</code> , and <code>Births</code> integer metrics to ensure structural accuracy before running transformations.
SQL Transformation & Aggregation	Develop and execute clean, well-commented SQL queries utilizing basic exploratory syntax (<code>SELECT *</code> and <code>LIMIT 1000</code>) to inspect the first comprehensive block of raw system records without active sorting overrides.
Insight Generation & Business Reporting	Compare raw structural layouts against standard relational database expectations to deliver definitive dataset validation findings for strategic leadership decisions.

This project does not include:

- **Total County Populations:** This analysis strictly measures raw birth volumes (`Births`) rather than calculating per capita birth rates against the total population of each county.
- **Gender Distinctions:** The scope isolates total infant delivery counts without filtering or segregating data based on the sex assigned at birth.
- **Stillborn and Mortality Metrics:** The dataset exclusively records live natality events; fetal deaths, stillbirths, and infant mortality tracking are completely excluded from this scope.
- Low-density population regional data categorization or sorting (reserved strictly for Phase 2 infrastructure tracks).
- High-density urban demographic sorting or hotspot isolation workflows (reserved strictly for Phase 3 optimization tracks).

Deliverables:

A specific list of things that this project will deliver.

Deliverable	Description/ Details
SQL Data Cleaning Script	A clean, production-ready, and well-commented <code>.sql</code> script demonstrating data ingestion, schema validation, and descending sorting parameters using BigQuery standard SQL syntax.
Analytical Dashboard	A highly visual dashboard focusing on regional birth concentrations across the major US geographic sectors (Northeast, South, Midwest, and West) to easily identify regional spikes and distribution anomalies.
Executive Summary Report	A brief presentation deck translating the top 10 high-priority county findings into clear resource allocation recommendations for healthcare stakeholders.

Schedule Overview / Major Milestones:

Project execution commences on June 16th, 2026, structured across highly focused technical milestones.

Milestone	Expected Completion Date	Description/Details
Milestone 1:	June 18th, 2026	Query Drafting & Ascending Constraint Validation — Completed within 24 Hours.
Milestone 2:	June 19th, 2026	System Extraction to Google Sheets & CSV Formatting — Completed within 48 Hours.
Milestone 3:	June 21st, 2026	Deployment of Dataset 02 to Local Project Directories — Completed within 72 Hours.

*Estimated date for completion:

*Final delivery, asset deployment, and repository verification are scheduled for **June 22nd, 2026**.*

The Analytical Imperative:

In healthcare logistics, data is directly tied to human outcomes. This analysis transcends simple numbers on a screen; by establishing a verified, unmanipulated structural baseline across national health data fields, it provides decision-makers with the accurate data framework required to model future clinical changes, evaluate record completeness, and ensure that maternal care resources are analyzed on a bulletproof foundation. True data maturity turns historical logs into protective, life-saving strategies.